OR-AS Operations Research Applications and Solutions	Case Name: **IJzertoren Memorial Square**	Sector	**Construction (civil)**	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions	
Date	**2015**	Project Control	Automatic tracking	
File Name	**C2015-33 IJzertoren Memorial Square**		Tracking based on user input	

1. Project description

Project authenticity

The IJzertoren Memorial Square project involves demolition, foundation work, street paving, and installation of street furniture and signage.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	12
Planned Duration (PD)	50 days*
Budget At Completion (BAC)	214 417.7 €
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	63%
Activity Distribution (AD)	57%
Length of Arcs (LA)	0%
Topological Float (TF)	14%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	20.8	19.6	-1.3
CRI-rho	19.6	19.7	-1.5
CRI-tau	12.3	13.6	-1.6

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	63.8	22.7	-1.0
SI	84.5	9.5	-1.2
SSI	23.2	16.1	-1.2
CRI-r	21.4	18.0	-0.9
CRI-rho	19.9	17.6	-1.0
CRI-tau	10.9	12.1	-1.1

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	7.1	4.1
PV - SPI	10.5	4.3
PV - SCI	22.3	3.3
ED - 1	7.0	3.9
ED - SPI	10.0	3.8
ED - SCI	16.6	2.4
ES - 1	5.8	3.5
ES - SPI(t)	10.0	3.7
ES - SCI(t)	17.2	2.0

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	5.6	-0.3
CPI	10.3	0.4
SPI	9.6	-0.1
SPI(t)	10.3	0.0
SCI	17.3	-1.5
SCI(t)	18.3	-1.7
0.8 CPI + 0.2 SPI	10.1	0.3
0.8 CPI + 0.2 SPI(t)	10.2	0.3

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 5 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-16268.4	-37284.4	-208.0	0.9	0.7	0.5	0.9
std dev	19373.3	38614.7	132.0	0.2	0.2	0.1	0.1
final	-10372.0	0.0	-399.0	1.0	1.0	0.5	1.0

2.3.3.2. Time forecasting

PD	50 days	Real Duration	100 days	Late	100%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	58.8	1.3	8.2	8.2
PV - SPI	59.4	0.9	8.1	8.1
PV - SCI	60.0	0.7	8.0	8.0
ED - 1	60.6	0.9	7.9	7.9
ED - SPI	61.2	1.3	7.8	7.8
ED - SCI	62.0	1.6	7.6	7.6
ES - 1	63.0	1.6	7.4	7.4
ES - SPI(t)	63.8	1.3	7.2	7.2
ES - SCI(t)	64.4	0.9	7.1	7.1

2.3.3.3. Cost forecasting

BAC	214 417.7 €	Real Cost	224 789.7 €	Over Budget	4.8%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	230686.1	19373.3	0.5	-0.5
CPI	243459.2	46972.5	1.7	-1.7
SPI	291336.1	67522.4	5.9	-5.9
SPI(t)	310817.5	78745.9	7.7	-7.7
SCI	311580.6	89822.6	7.7	-7.7
SCI(t)	332254.1	97683.9	9.6	-9.6
0.8 CPI + 0.2 SPI	249294.3	45047.7	2.2	-2.2
0.8 CPI + 0.2 SPI(t)	251232.9	45195.8	2.4	-2.4